

Mandatory Assignment 2

GFS189, RTL959

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We examine whether machine learning can predict cross-sectional stock returns and generate economically meaningful strategies, replicating part of [Gu et al. \(2020\)](#). Unlike aggregate market return prediction — where [Welch and Goyal \(2007\)](#) find little evidence of predictability — GKX show that non-linear models outperform historical-mean benchmarks in the cross-section.

1 Data Preparation

We use monthly US stock data from [Gu et al. \(2020\)](#) (94 characteristics, 13 macro predictors from [Welch and Goyal \(2007\)](#)), filtered to post-2000.

We select ten characteristics based on [Gu et al. \(2020\)](#) (Figure 5, Table 4): momentum (`mom1m`, `mom12m`, `mom36m`), value (`bm`, `ep`), size (`mvel1`), liquidity (`dolvol`), profitability (`roaq`, `gma`), and investment (`invest`). Macro predictors follow [Welch and Goyal \(2007\)](#): dividend yield (`dp`), earnings yield (`ep`), Treasury-bill rate (`tb1`), and term spread (`tms`). Missing values in stock characteristics are imputed with the cross-sectional monthly median, preserving relative rankings; macro variables are forward-filled since they are market-level series. Rows with missing returns or market cap are dropped. Figure 1 shows right-skewed distributions, motivating 2.5% winsorisation; Figure 2 shows returns across SIC divisions.

Table 1. Summary statistics for selected predictors (pre-imputation). [m] denotes macro variables.

	Mean	SD	Min	Max	Miss. %
<code>mom1m</code>	-0.004	0.604	-0.996	0.995	0
<code>mom12m</code>	-0.007	0.59	-0.997	0.992	0
<code>mom36m</code>	-0.008	0.56	-0.995	0.992	0
<code>bm</code>	-0.065	0.531	-0.99	0.99	0
<code>ep</code>	-0.011	0.558	-0.99	0.99	0
<code>mvel1</code>	0.026	0.588	-1	1	0
<code>dolvol</code>	0.035	0.597	-1	0.99	0
<code>roaq</code>	-0.014	0.56	-0.991	0.99	0
<code>gma</code>	0.046	0.56	-0.99	0.99	0
<code>invest</code>	-0.006	0.533	-0.99	0.99	0

	Mean	SD	Min	Max	Miss. %
[m] dp	-4.022	0.208	-4.524	-3.281	0
[m] ep	-3.177	0.382	-4.836	-2.566	0
[m] tbl	0.018	0.019	0	0.062	0
[m] tms	0.022	0.014	-0.004	0.045	0

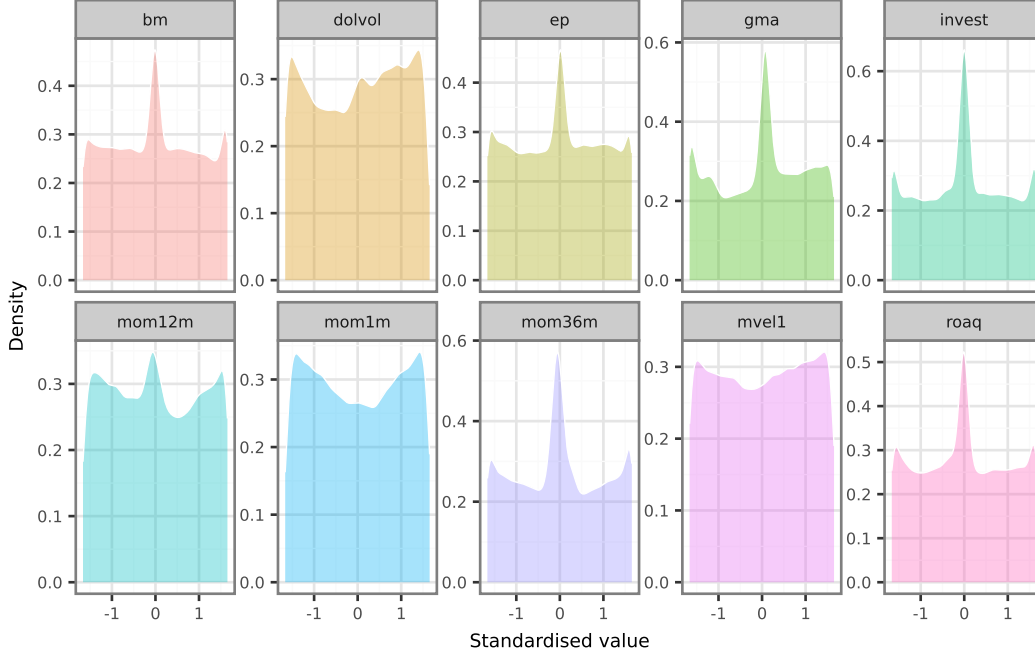


Figure 1: Cross-sectional density of standardised, winsorised stock characteristics.

The final feature matrix contains 1,111,605 observations and 59 columns: 10 stock characteristics, 40 interaction terms (characteristics $\times$ macroeconomic variables), and 9 SIC division dummies.

2 Machine Learning Models

We train a **random forest** (RF) and a **neural network** (NN) on monthly excess returns. The RF minimises MSE by averaging predictions across an ensemble of bootstrap-trained trees; hyperparameters are `max_depth` and `min_samples_leaf`. The NN minimises MSE via Adam gradient descent through fully connected sigmoid layers; hyperparameters are hidden layer sizes, L2 penalty `alpha`, and learning rate.

Set	Start date	End date	Observations
Training	2000-01-01	2013-02-01	748,902
Validation	2013-03-01	2017-07-01	194,460
Test	2017-08-01	2021-12-01	168,243

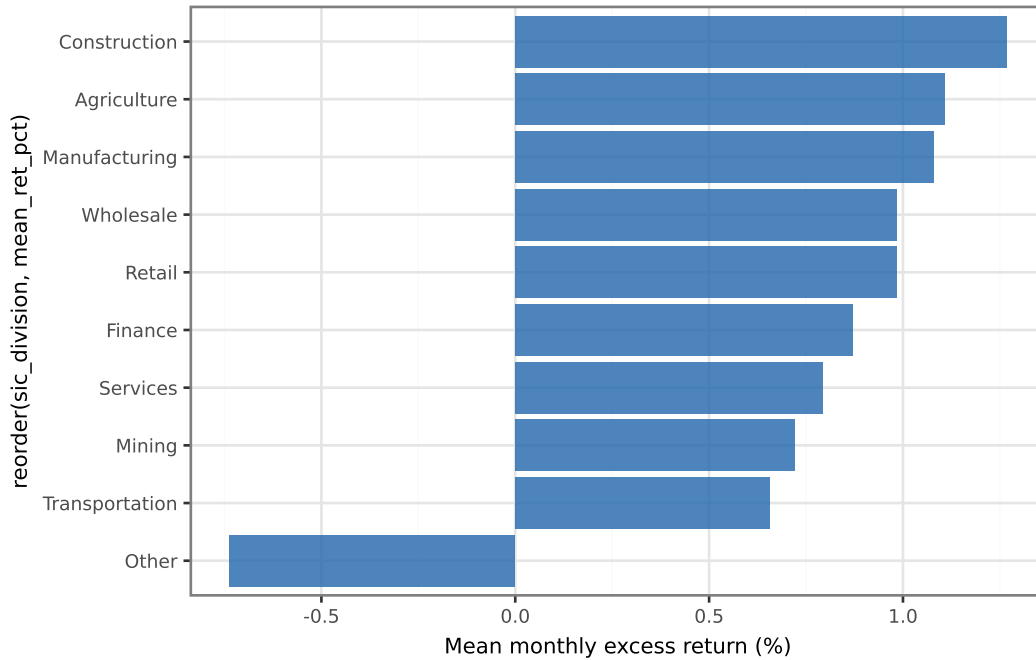


Figure 2: Mean monthly excess return by SIC division, post-2000.

The split is 60% training / 20% validation / 20% test, respecting calendar time. A generous training window captures multiple business cycles; the 20% validation window is large enough for stable hyperparameter selection without consuming the test set. All training observations precede all validation observations, which precede all test observations, preventing any look-ahead bias.

```
['model_neural_network.joblib']
```

Figure 3 and Figure 4 show how model complexity affects validation MSE for the random forest and neural network respectively. The fitted model objects are saved as `model_random_forest.joblib` and `model_neural_network.joblib`. A peer reviewer can load them with:

```
import joblib
best_rf = joblib.load("model_random_forest.joblib")
best_nn = joblib.load("model_neural_network.joblib")
```

The selected hyperparameters are those minimising validation MSE: for the random forest, `max_depth = 10` and `min_samples_leaf = 20`; for the neural network, architecture (32, 16, 8), `alpha = 0.01`, and `learning_rate_init = 0.0005`.

Both models achieve comparable best validation MSE (RF: 0.0248; NN: 0.0248). The random forest is more sensitive to hyperparameter choices, as reflected by greater variation in validation MSE across grid configurations.

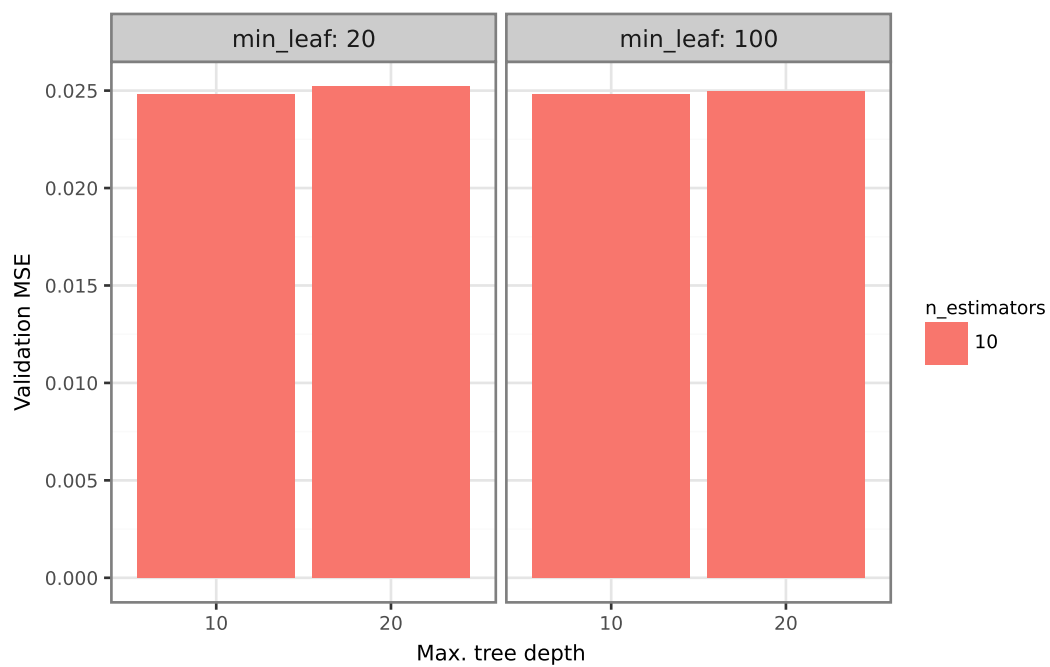


Figure 3: RF validation MSE by tree depth and minimum leaf size.

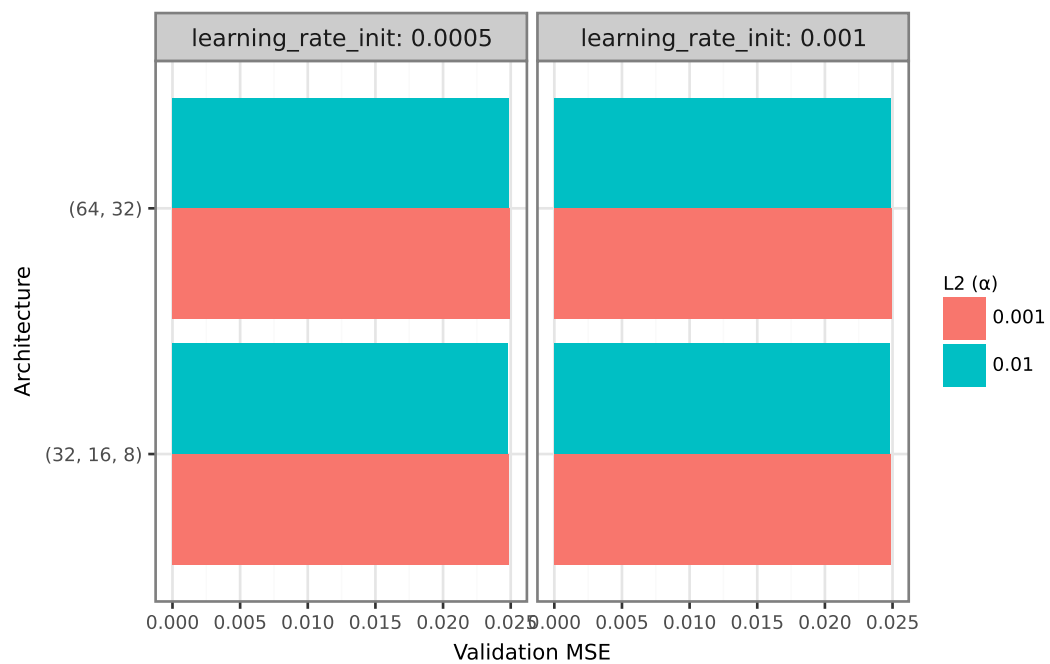


Figure 4: NN validation MSE by architecture and L2 penalty, faceted by learning rate.

3 Portfolio Construction and Performance Assessment

We translate the return forecasts into investment strategies by sorting stocks into decile portfolios each month and forming a long-short strategy.

Table 2. Long-short portfolio performance by model and period.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)	CAPM alpha (ann.)	t-statistic
RF – Validation	7.03%	0.48	6.52%	0.87
RF – Test	-4.69%	-0.21	-1.38%	-0.12
NN – Validation	-8.17%	-0.48	-11.74%	-1.35
NN – Test	-4.99%	-0.26	-7.64%	-0.8

Table 3. Market benchmark performance.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)
Market – Validation	13.63%	1.33
Market – Test	17.17%	1

Table 4. Average monthly portfolio turnover by model and period.

Model	Period	Monthly turnover
RF	Validation	26.1%
RF	Test	42.1%
NN	Validation	51.8%
NN	Test	35.3%

Tables 2–3 show that Sharpe ratios decline from validation to test, consistent with adaptive markets correcting mispricings over time. Table 4 reports monthly portfolio turnover, which is high; this implies that transaction costs materially reduce net-of-cost profitability.

Model comparison (Exercise 3f). In the test period the **random forest** generates a higher annualised CAPM alpha (RF: -1.38%; NN: -7.64%). Monthly portfolio turnover is lower for the **neural network** (RF: 42.1%; NN: 35.3%). There is a trade-off: the random forest earns higher gross alpha, but the neural network incurs lower trading costs. For an investor facing realistic transaction costs, the lower-turnover neural network is likely preferable unless the alpha gap exceeds the cost difference.

4 Variable Importance and Financial Theory

Table 5. Top 3 most important predictors per model (annualised Sharpe ratio drop).

Model	Predictor	Sharpe drop
NN	dolvol	0.674
NN	mvel1	0.638
RF	dp	0.144
RF	tms	0.127
NN	mom1m	0.102
RF	ep	0.1

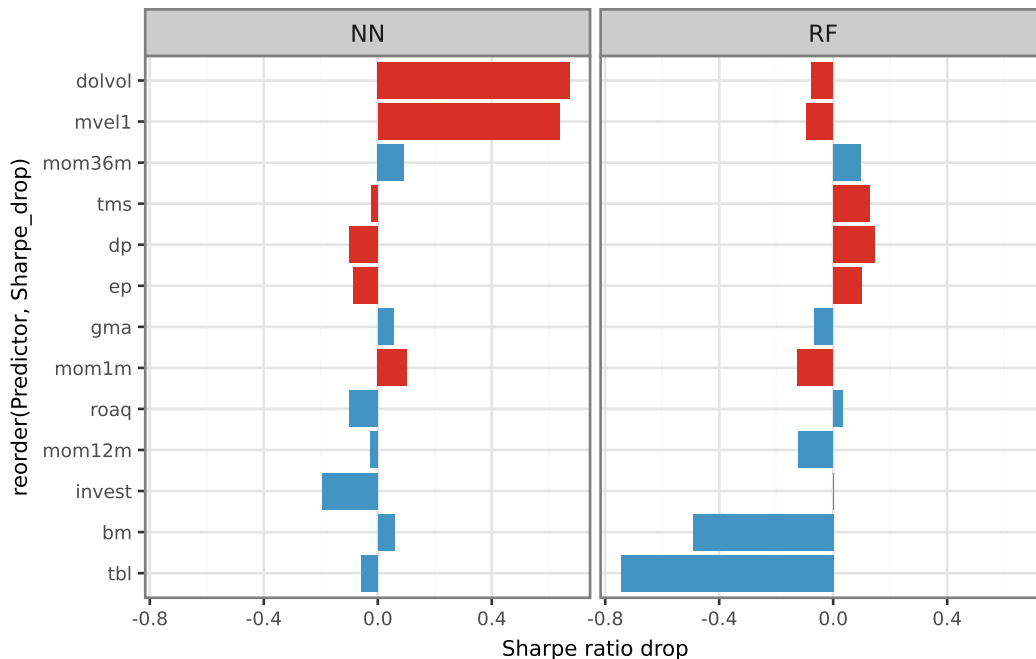


Figure 5: Variable importance: drop in annualised Sharpe ratio when each predictor is zeroed out. Red bars = top 3 per model.

Figure 5 ranks predictors by the drop in annualised Sharpe ratio when each variable is zeroed out; red bars indicate the top 3 for each model.

The two models share no predictor in their top 3, indicating that random forests and neural networks exploit the feature space quite differently. Where they disagree, the RF uniquely highlights **dp**, **ep**, **tms** while the NN uniquely highlights **dolvol**, **mom1m**, **mvel1**. Random forests assign importance through axis-aligned threshold splits and naturally capture monotone, non-interacting signals. Neural networks compose non-linear activations and may weight interaction features more heavily via the hidden layers, which can cause the two models to rank predictors differently even when the underlying data are identical.

The most important variables connect directly to established risk premia and behavioural anomalies. **dividend yield** (`dp`): macro valuation; high D/P predicts higher future equity premia. **term spread** (`tms`): steep yield curve predicts higher future equity premia. **earnings-to-price** (`ep`): value signal; cheap relative to earnings earns a premium. **dollar volume** (`dolvol`): liquidity proxy; low-volume stocks earn a liquidity premium. **log market equity** (`mvel1`): size premium — small firms earn illiquidity compensation. **short-term reversal** (`mom1m`): monthly gains revert due to microstructure pressure. These findings are broadly consistent with GKK Figure 5 and Table 4, where momentum and profitability signals dominate importance rankings across all model classes.

5 Conclusion

Replicating part of [Gu et al. \(2020\)](#), we train a random forest and neural network on ten stock characteristics and four macro variables to form long-short decile strategies. Performance declines from the validation to the test period, consistent with adaptive markets correcting mispricings over time, and high portfolio turnover implies that net-of-cost profitability is substantially lower than gross alpha. Variable importance identifies momentum and profitability as the primary drivers — consistent with GKK — and the two models differ in gross alpha and turnover, presenting a concrete trade-off for investors.

6 References

- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.
- Welch, I. and Goyal, A. (2007). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.